

Question 1 (2 pts)

Which of the following statements about methods for addressing overfitting in deep neural networks is incorrect?

- A) Data augmentation reduces overfitting by diversifying the training distribution, but applying identical augmentations to the validation set may bias generalization metrics.
- B) Early stopping typically monitors validation loss, but in some cases, tracking training loss can prevent overfitting when validation data is limited.
- C) Batch Normalization provides a regularization effect through batch-dependent stochasticity, though its primary role is to stabilize training by normalizing intermediate activations.
- D) Weight decay, implemented as L2 regularization, promotes sparsity in the network by shrinking weights, thereby reducing model complexity and overfitting.

Write your **CAPITAL** letter answer in this box:

Question 2 (2 pts)

Deep learning involves lots of matrix multiplication. Given three matrixes $A \in R^{m \times n}$, $B \in R^{n \times p}$ and $C \in R^{p \times q}$ with $m < n < p < q$, which of the following is the most efficient calculation?

- A) $(AB)C$
- B) $A(BC)$ if C is highly sparse, otherwise $(AB)C$
- C) $A(BC)$
- D) All above are the same.

Write your **CAPITAL** letter answer in this box:

Question 3 (15 pts)

(Diffusion Model was not covered as a core topic in that year's course syllabus. The teaching focus of the generative models section was on GANs and VAEs, while diffusion models were introduced only at a general level in lectures, as a cutting-edge Research Extension. For this reason, the question provides sufficient background information in the parts concerning diffusion models. The intent is to require students to draw on the given background, combine it with the GAN/VAE and other generative model knowledge taught as core content in class, and arrive at the correct answer through induction and reasoning.

It is worth adding that, for the current year, diffusion models have been formally included as one of the core teaching topics of the course. Accordingly, if diffusion models appear in this year's assessment, it is likely that such extensive background information will no longer be provided; instead, students will be expected to answer directly on the basis of the relevant knowledge they have acquired.)

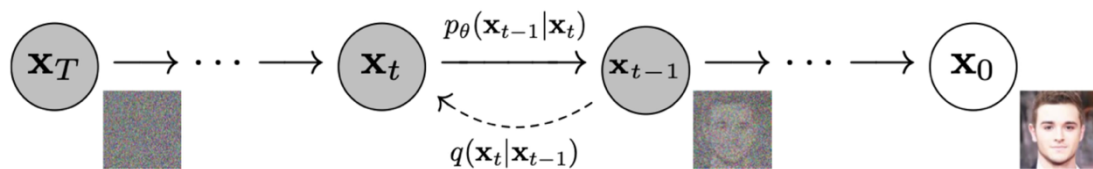


Figure 1. Diffusion Model

In a digital art creation platform, a generative model produces high-resolution 512x512 images (e.g., landscapes, portraits) from random noise, responding to user text prompts like “sunset over mountains.” Traditional models, such as GANs and VAEs, are hindered by some limitations. In contrast, Diffusion Models have gained popularity by effectively addressing their issues.

A Diffusion Model adds incremental Gaussian noise to an image over T steps (e.g., $T=1000$), transforming it into pure noise via a Markov chain. A neural network learns to reverse this process, iteratively refining noisy data into a clear image by predicting noise removal at each step. The model is trained to minimize the distance between predicted and actual noise, ensuring realistic image reconstruction. During inference, it starts from random noise and applies T iterative steps, which is computationally intensive (e.g., 10 seconds per image). Challenges include generating images matching diverse prompts, reducing inference time and maintaining quality on limited training data.

1. The training objective function used in the diffusion model is defined similarly to the reconstruction loss used in VAE, minimizing the distance between predicted and actual noise. Assume that a diffusion model receives input: latent variable x_t) and timesteps t , gives a predicted noise $\epsilon_\theta(x_t, t)$, the actual noise is defined as ϵ , define the reconstruction loss used in diffusion model. **(3 pts)**
2. To reduce the Diffusion Model's inference time while maintaining quality, propose two techniques inspired by GAN, VAE or general deep learning principles, and explain how these techniques address the challenge. **(6 pts)**
3. Identify two key limitations of GANs or VAEs, and explain how the mechanism of Diffusion Models avoids or overcomes those limitations. **(6 pts)**

Question 4 (15 pts)

(In that year, Mamba was not part of the lecture syllabus, which covered only models such as RNN/LSTM and Transformer. This is a novel-context discussion question: all necessary background is provided in the stem, and each part targets a specific concept taught in lectures — here, the positional handling, long-range dependency, and memory-bottleneck behaviour of LSTM/Transformer. The stem specifies Mamba's algorithmic structure on these same aspects, and students must combine their prior knowledge with the given information to produce an integrated analysis. Such questions are the capstone (most challenging) part of the final exam.)

This question introduces a new model but is fully answerable based on the description and your knowledge of RNNs and Transformers.

Recent advances in sequence modeling have led to the development of a new architecture called Mamba, which belongs to a family of models known as State Space Models (SSMs). Like RNNs, Mamba processes sequences in an autoregressive, step-by-step manner, but unlike classical RNNs, it introduces linear recurrence with learnable dynamics that can be computed in parallel.

A simplified view of Mamba can be described as follows:

- Mamba maintains a fixed-size hidden state vector at each time step (similar to RNNs), which is meant to summarise all past inputs seen so far.
- However, the state update is defined not by a nonlinear recurrence (like tanh or GRU), but through a parameterized linear dynamical system:

$$s_t = A \cdot s_{t-1} + B \cdot x_t,$$

where A and B are learnable matrices, x_t and s_t are the input and the update state at timestep t .

- To enhance performance, Mamba includes a data-dependent gating mechanism, which controls how much of the new input x is integrated into the current state.
- Unlike RNNs, Mamba's design allows the entire sequence of hidden states to be computed in parallel, using a fast scan algorithm, even though the model is autoregressive.
- Unlike attention-based models such as Transformers—which keep and revisit the representations of all previous tokens, so that their per-step cost and memory grow with the sequence length—Mamba carries information forward only through this single fixed-size state.
- Mamba has shown strong performance on long-sequence tasks such as document classification, long-range time-series modeling, and speech recognition, where Transformers often struggle with quadratic attention costs.

1. A standard Transformer, without any extra component, cannot tell the order of the input tokens, so it must rely on positional encoding. Which property of self-attention causes this problem? In contrast, why does the way Mamba processes a sequence allow it to work without positional encoding? **(4 pts)**
2. When an LSTM processes a very long sequence, information from far-away steps is often hard to keep until the current step. What is the root cause of this difficulty? Through what different mechanism does a Transformer avoid it, so that information from distant tokens barely fades? And on this point, how is Mamba similar to, and different from, the LSTM and the Transformer? **(5 pts)**
3. A Transformer can access all past tokens without any loss, but this ability comes at a cost. When generating a very long sequence, what bottleneck does a Transformer face, and what is its root cause? What different design do Mamba (and RNNs) use to avoid this bottleneck? And what new limitation does this design itself bring? **(6 pts)**